

Ranking Transfers, Thresholds Don’t: Prior-Corrected Test-Time Adaptation for Audio Deepfake Detection

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Abstract—Audio deepfake detectors generalise poorly across corpora despite near-perfect in-domain accuracy. We show the failure is one of *operating point*: a fine-tuned self-supervised detector’s *ranking* (ROC-AUC) largely transfers cross-corpus while its decision *threshold* does not, so unlabeled target audio can repair the operating point at test time. We further show that confident-quantile test-time adaptation (TTA) silently assumes a *class-balanced* target pool — an assumption that holds where TTA is measured and essentially nowhere else, in either direction: the standard ASVspoof2021-DF benchmark is 97.2% spoof, In-the-Wild is 37.2%. On the official DF eval (400,435 trials) the published symmetric-quantile configuration destroys the operating point of three independently trained state-of-the-art released checkpoints, driving accuracy from 98.8–99.1% to 62–65%; three of four modern TTA baselines (Tent, SHOT, ETA) degrade them too, each through its own version of the same assumption, while SAR avoids damage only by reverting its own updates. Correcting the pseudo-label budget with a label-free estimate of the target prior ($\hat{\pi}=0.9715$ vs. a true 0.9708) instead preserves accuracy and improves EER on all three, beating the symmetric budget by 0.79–1.82 points and the source model by up to 27.9% relative. Because EER and AUC both move, no re-thresholding rule accounts for it: the one-line median-threshold control that looks competitive on balanced pools loses 40 accuracy points at this benchmark’s real prior. Two label-free monitors — rank agreement with the frozen source model, and prediction-rate drift — separate the adaptations that damaged a checkpoint from those that did not.

Index Terms—audio deepfake detection, anti-spoofing, test-time adaptation, label shift, cross-corpus generalisation, EER.

I. INTRODUCTION

Self-supervised (SSL) front-ends such as wav2vec 2.0 and XLS-R, paired with lightweight back-ends, reach very low in-domain error on standard anti-spoofing benchmarks [1]–[3]. Deployed on audio from a different source—new microphones, codecs, languages, or generators—accuracy drops steeply, a well-documented generalisation gap [5], [6], [17].

We start from a diagnostic measurement: a fine-tuned XLS-R detector evaluated on an unseen corpus keeps a high **ROC-AUC while its accuracy at a fixed threshold collapses**. It still *ranks* clips correctly; its operating point is mis-calibrated. Because ranking transfers, the most-confident predictions on unlabeled target audio are trustworthy enough to adapt on at test time.

That observation is the starting point of a family of methods, not a method. The step every member of the family takes—label the extreme quantiles of the target scores and self-train on them—carries an assumption that is invisible on the benchmarks the field uses and fatal on the pools it deploys to: that the target pool is roughly class-balanced. Take the top and bottom 30% of a 97%-spoof pool and the “confidently bona fide” bucket is mechanically made of low-scoring *spoof*. Self-training then teaches the detector that a great deal of spoof is genuine. This is not hypothetical: it is what happens to a released state-of-the-art checkpoint on the official ASVspoof2021-DF protocol (Table I).

Contributions. (i) The ranking–calibration asymmetry as the mechanism of cross-corpus failure, over 369 third-party cells and tested directly: a proxy $\mathcal{A}$ -distance *rises* after adaptation, so the effect is operating-point repair, not learned invariance. (ii) The class-prior assumption hidden in confident-quantile TTA and in three of four modern TTA baselines, measured on the field’s standard benchmark at its real prevalence, where it destroys the operating point of three released SOTA checkpoints. (iii) A label-free correction—shift estimation of the target prior, used to split the pseudo-label budget—which removes the damage and beats the symmetric budget on every checkpoint, threshold-free. (iv) Two label-free monitors that separate damaging adaptations from safe ones, including our own published configuration.

Related work. SSL front-ends with AASIST or attentive back-ends dominate recent benchmarks [1]–[4]. Train-time domain generalisation aligns source domains [5] or runs gradient reversal over source and target jointly [13]; both act before deployment. In TTA, entropy minimisation (Tent) [7] is the reference recipe, and its instability prompted reliable-sample filtering (ETA/EATA) [8], sharpness-aware entropy with a recovery scheme (SAR) [9], and information maximisation with self-labelling (SHOT) [10] — we compare against all four. Batch-statistics recalibration [14] is a no-op for LayerNorm. The prior estimator is black-box shift estimation [11]; the proxy $\mathcal{A}$ -distance is from [15].

II. METHOD

Let x be a waveform with label $y \in \{0, 1\}$ (fake= 1) and let f_θ produce $s_\theta(x) = P_\theta(y=1 \mid x)$. We are given a source

model, its labelled training corpus $\mathcal{S}$, and an *unlabelled* target pool $\mathcal{T}$. No target label is read at any point.

What breaks under corpus shift. Ranking quality $\text{AUC} = \Pr[s_\theta(x^+) > s_\theta(x^-)]$ is threshold-free; accuracy at a fixed τ is not. Empirically AUC degrades far less than accuracy at $\tau=0.5$ under source→target shift: across 369 third-party evaluation cells the accuracy a model’s ranking supports but its shipped threshold does not deliver averages 7.29 points. The deployable object is therefore the score *order*, and the broken object is the operating point. Everything below follows from taking that literally.

(a) The target prior, label-free. An operating point is a statement about a prior, so we estimate the prior rather than assume it. Black-box shift estimation [11] gives $\hat{p} = M^{-\top} q$, where $M[y, \hat{y}] = P_S(\hat{y} | y)$ is the source confusion matrix measured on the model’s own labelled training data and q is the observed target prediction histogram. Label-free prior estimators divide into two families, and the split matters more than we expected. BBSE and EM [12] read the model’s *thresholded prediction rate* on the target; a two-component Gaussian mixture reads the *shape* of the target score histogram. The first family inherits whatever calibration error the target induces — exactly the quantity this paper says does not transfer. On In-the-Wild, where the checkpoint predicts 58.9% fake on a 37.2%-fake pool, BBSE returns $\hat{\pi}=0.600$ and EM 0.614 while the mixture returns 0.422; on the well-calibrated DF pool all three agree to within 0.02. A prediction-rate estimator is accurate where the operating point is already sound and unreliable where it is not. We report DF with BBSE as run (a mixture prior gives a near-identical budget there); on a badly mis-calibrated pool the mixture is the safer default. What matters most is estimating the prior at all rather than assuming it is $\frac{1}{2}$.

(b) Prior-corrected confident tails. The standard rule takes the bottom q and top q of the target scores as pseudo-real and pseudo-fake. We split the same total budget $2q$ by the estimated prior,

$$q_{\text{real}} = 2q \hat{\pi}_{\text{real}}, \quad q_{\text{fake}} = 2q (1 - \hat{\pi}_{\text{real}}), \quad (1)$$

which is *identical* to the symmetric rule at $\hat{\pi}_{\text{real}} = 0.5$ — so on a balanced pool the correction is provably inert and cannot be the source of any balanced-pool result — and shrinks the pseudo-real bucket to the estimated real fraction otherwise. The prior is estimated *once*, from the frozen pre-adaptation model: re-estimating each epoch lets a drifting model reinforce its own mistakes.

The obvious prior-free alternative is a fixed *confidence* threshold ($p \geq 0.95 \Rightarrow \text{fake}$) instead of a quantile. It does not remove the problem, it moves it: a quantile rule is calibration-free but assumes the prior, a confidence rule is prior-free but assumes calibration — and calibration is exactly the property the first paragraph of this section measures as *not* transferring across corpora. We run it as a baseline (Table I) rather than argue about it.

(c) Objective. With pseudo-labels $\tilde{y}$ from (b) on the confident set $\mathcal{C}$ and a stochastic channel perturbation $a(\cdot)$ (per-clip gain $g \sim \mathcal{U}(0.7, 1.3)$ plus Gaussian noise $\sigma=0.005$),

$$\min_{\{\gamma, \beta\}, \phi} \underbrace{\sum_{x \in \mathcal{C}} \text{CE}(f_\theta(x), \tilde{y})}_{\mathcal{L}_{\text{ST}}} + \lambda \underbrace{\|f_\theta(a(x)) - \text{sg}[f_\theta(x)]\|^2}_{\mathcal{L}_{\text{cons}}}, \quad (2)$$

optimising only the LayerNorm affine parameters $\{\gamma, \beta\}$ of the fine-tuned blocks and the head ϕ (16,706 parameters), with $\lambda=0.3$, $q=0.3$. Restricting the trainable set and using only prior-corrected confident tails is what prevents the degenerate collapse to which entropy minimisation is prone.

(d) A label-free guard. Adaptation needs a stop, and the step count has been the weakest-specified part of TTA recipes here. Since the source ranking is the asset being exploited, we monitor rank agreement with the frozen source model, $\rho_t = \text{Spearman}(s_{\theta_t}(\mathcal{T}), s_{\theta_0}(\mathcal{T}))$, and keep the last step with $\rho_t \geq \tau$ (Sec. IV).

III. EXPERIMENTAL SETUP

Protocol A (published baselines exist). The official ASVspoof2021-DF eval partition against the official CM keys; 400,435 of 611,829 trials are held locally, the missing part uniform across codec, label and phase (verified 75–25 in every stratum), true spoof rate 97.22%. For scale, published pooled EERs on this protocol run from 1.95 (Wav2DF-TSL) and 2.87 (SSL+SLS) through 8.54 (Wav2Vec2-AASIST) to 15.64 (challenge DF top-1); our own source model reaches only 26.33, so we run the method on *released* checkpoints instead: DeepFense XLS-R 300M + AASIST trained on ASVspoof2019-LA [22], at three independent training seeds. Adaptation uses an unlabelled 20,000-clip subsample of the eval pool; we report the full official pooled EER and, as an inductive check, the EER over trials *not* in that subsample.

Protocol B (cross-corpus, balanced). Leave-one-corpus-out over four EER-capable targets: ASVspoof2019-LA [18]; D2 (LibriSpeech bona fide vs. eight open TTS systems); In-the-Wild [17]; and the Arabic Audio Deepfake set [20]. Each held-out target’s source pool merges the other three with a 38-language MLAAD subset [19]; held-out target audio is never in any source pool and six MLAAD languages are held out of every pool. Ten seeds per target per method.

Protocol C (a benchmark at its own prevalence). All 31,779 In-the-Wild clips at their native 37.2% fake rate — skewed the *opposite* way to DF’s 97.2%.

Comparison points. Source-only; the median-threshold rule; Tent, SHOT, ETA/EATA, SAR, AdaBN [14]; DANN [13] (trained to convergence with the held-out target’s unlabeled inputs — more privileged access than ours); an ASDG-style baseline [5]; and a target-supervised oracle. Every TTA arm adapts the same 16,706 parameters with the same lr, batch and pass count; none was tuned separately, and none could be, since per-target TTA tuning needs the target labels the setting assumes absent. Two published constants in the modern TTA baselines are degenerate at $C=2$ and were re-derived rather than transplanted. EATA’s redundancy filter ($|\cos(p, \bar{p})| < 0.05$)

TABLE I

OFFICIAL ASVSPOOF2021-DF EVAL, 400,435 TRIALS, 97.22% SPOOF, ON A *released* XLS-R+AASIST CHECKPOINT [22] WE DID NOT TRAIN; ALL ADAPTATION IS LABEL-FREE. DEFICIT $\delta = (100 - \text{EER}) - \text{acc}@0.5$ IS THE ACCURACY THE RANKING SUPPORTS BUT THE THRESHOLD DOES NOT DELIVER. ρ AND THE PREDICTION-RATE GAP ARE THE LABEL-FREE MONITORS OF SEC. II(D) — COMPUTABLE AT DEPLOYMENT, UNLIKE EVERY OTHER COLUMN.

	EER %	AUC	acc@0.5	deficit	ρ	gap
Source (as released)	4.51	.9923	98.87	−3.38	1.000	.000
+ median-threshold rule	4.51 [†]	.9923 [†]	59.06	—	—	—
+ ETA [8]	7.44 [‡]	.9288	98.93	−6.37	0.312	.007
+ Tent [7]	4.06 [‡]	.9765	98.52	−2.58	0.438	.013
+ SHOT [10]	6.38	.9725	77.64	+15.99	0.767	.225
+ TTA, fixed confidence 0.95	[8b]					
+ TTA, symmetric $q=0.3$	5.84	.9846	62.96	+31.20	0.816	.372
+ TTA, prior-corrected	4.02	.9931	98.95	−2.97	0.867	.007
(disjoint/inductive half)	4.03	.9931	98.95			
+ SAR [9]	4.47 [§]	.9923	98.88	−3.35	0.970	.000

[†] A threshold cannot change EER or AUC; both come from the score ROC. The median rule is a control for the *accuracy* claim and cannot produce any EER result in this paper. [‡] EER over a pool that is more than 93% a single tied score (Tent saturates 98.07% of scores to exactly 1.0, ETA 97.71%); read AUC. [§] SAR neither damages nor adapts: 308 recovery resets in 2,500 steps, AUC unchanged to four decimals.

is calibrated for $C=1000$; on the binary simplex the minimum attainable cosine is 0.71 at $\bar{p}=(.5, .5)$ and 0.11 at $(.9, .1)$, so nothing passes until the running average itself reaches a vertex, after which only *minority*-predicted samples pass — neither is the stated mechanism, so we run the reliability filter alone, which is EATA’s own published ablation. SAR’s recovery trigger $e_0=0.2$ sits far below ImageNet’s $E_0=0.4 \ln 1000=2.76$ but *inside* the $C=2$ reliable band $H<0.277$, firing on every batch and freezing the model at its initial weights (verified: 32 resets in 32 steps, zero net movement); we keep its relative position, $e_0 = (0.2 / \ln 1000) \ln C = 0.020$.

Implementation. 16 kHz audio, top-4-layer XLS-R fine-tuning for our own source models (Adam, $\text{lr} \times 10^{-4}$, batch 32, 8 epochs). Adaptation: $\text{lr} \times 10^{-4}$, $E=4$ passes, batch 32 (third-party crops matched to each checkpoint’s own training length). Code and manifests: [URL withheld].

IV. RESULTS

A. The class prior is the hidden variable

Table I is the main result. Applied unmodified to a released checkpoint that is already better than the published Wav2Vec2-AASIST baseline on this protocol, the symmetric-quantile configuration *damages* it: accuracy $98.87 \rightarrow 62.96\%$, EER $4.51 \rightarrow 5.84\%$, and it manufactures a 31-point calibration deficit where there was none. The mechanism is mechanical: $q=0.3$ labels the bottom 30% of a 97.2%-spoof pool “confidently bona fide”.

The published baselines are no better placed, and each fails through its own version of the assumption. SHOT minimises $\mathbb{E}[H(p)] - H(\mathbb{E}[p])$, whose diversity term is maximised when the mean prediction is *uniform* — at $C=2$ a hard-coded prior of $\frac{1}{2}$, asking it to move 47 points of probability mass into the bona fide class: the regulariser introduced to *cure* Tent-style collapse encodes the assumption more literally than Tent does. ETA fails more instructively still. Its reliability filter exists to discard the high-entropy samples that destabilise

TABLE II

THE COMPLETE IN-THE-WILD BENCHMARK (31,779 CLIPS, PRIOR 0.372), FOUR RELEASED CHECKPOINTS ORDERED BY SOURCE CALIBRATION DEFICIT δ . EER % / ACCURACY %. THE SYMMETRIC BUDGET REPAIRS EVERY CHECKPOINT THAT HAS A DEFICIT; THE PRIOR-CORRECTED ONE DOES NOT, AND ITS SHORTFALL GROWS WITH δ — WHICH IS WHAT CORRUPTS THE PRIOR ESTIMATE (SEC. V).

checkpoint	δ_{src}	source	symmetric q	prior-corrected
DeepFense s2	2.1	9.64 / 88.3	9.71 / 85.7	9.89 / 87.1
ssl-aasist [21]	16.7	5.06 / 78.2	4.55 / 95.4	4.54 / 95.2
DeepFense s240	22.6	8.47 / 68.9	8.33 / 88.8	8.48 / 77.0
DeepFense s42	44.4	16.47 / 39.1	12.58 / 82.4	14.51 / 48.0

entropy minimisation; here it admitted 99.8% of them, so ETA reduced to Tent and landed worse. That is structural: an entropy-keyed filter separates confident from uncertain, not *confident-and-right* from *confident-and-wrong*, and mis-calibration manufactures exactly the latter — a safeguard keyed on the model’s own confidence inherits the broken calibration it guards against.

SAR is the exception that clarifies the rule. It is the only prior method here that does not degrade the checkpoint — and it manages that by declining to move it: 308 model-recovery resets in 2,500 steps, an AUC unchanged to four decimals, and a score distribution indistinguishable from the source model’s. Its collapse guard does exactly what it promises, and prevention is all it delivers. So the claim is not that every published method fails here. It is that *three of the five prior configurations degrade a state-of-the-art detector at the standard benchmark’s own prevalence, the fourth avoids damage only by reverting its own updates, and none of them improves it.*

Eq. (1) with a label-free prior estimate reverses this: BBSE returns $\hat{\pi}=0.9715$ against a true 0.9708, the budget becomes (0.017, 0.583), and EER improves to 4.02%. The change between the two arms is one scalar. Over three independently trained checkpoints (EER % / accuracy %):

	seed 2	seed 42	seed 240
source	4.51 / 98.87	4.56 / 98.78	3.82 / 99.11
+ symmetric q	5.84 / 62.96	4.08 / 61.80	4.66 / 65.24
+ prior-corrected	4.02 / 98.95	3.29 / 98.98	3.80 / 98.74

Two things replicate and one does not. The *operating-point* collapse under a symmetric budget holds on all three (accuracy 62–65%, $\delta+30$ to $+34$), as does prior-correction beating that budget (−1.82, −0.79, −0.86 EER). What does not replicate is the symmetric budget’s effect on *ranking* — EER worse on two, better on one. A mis-specified prior reliably breaks the operating point, not the ranking. Gains over source-only vary widely (1.27, 0.49, 0.02 EER), so we rest on the unanimous controlled comparison.

B. The failure, and where the recipe already works

Table II measures the opening claim at a *mild* prior (0.372). DeepFense s42 — the same checkpoint that scores 4.56% on DF — arrives on In-the-Wild ranking at AUC 0.917 while being correct on 39.1% of clips, fewer than a majority-class predictor. Confident-quantile self-training repairs exactly that:

on the three checkpoints with a deficit it drives δ from 16.7, 22.6, 44.4 to 0.1, 2.9, 5.0, lifting accuracy by up to 43 points, while on the well-calibrated one it has nothing to repair and costs a little. On s42 it also raises AUC .917 $\rightarrow$.947 — which no threshold can do — and the disjoint half is slightly better (12.27 / .949), so the ranking gain is not memorisation.

C. Why this is not a re-thresholding result

Both EER and AUC improve, and neither takes a threshold as input, so no operating-point rule can produce this gain. The cleanest form of that argument uses the *same* prior estimate twice: re-point the threshold with it and take no gradient step, and accuracy moves 98.92 $\rightarrow$ 98.94% while EER stays at 4.51 by construction; feed it to the pseudo-label budget instead and EER goes to 4.02. The prior does its work in the budget, not at the threshold.

On *balanced* pools a one-line median-threshold rule is genuinely competitive with gradient adaptation on accuracy — we reported it as such, and it is the right control for a calibration claim. Resampling the DF scores across target priors shows why, and what it costs: the median rule gives 95.6% accuracy at a prior of 0.5 (shipped threshold 88.3, labelled oracle 95.7) and falls monotonically to 59.0% at the benchmark’s own 0.972, while a threshold at the $(1-\hat{\pi})$ quantile tracks the shipped one throughout.

That is the rule’s assumption, not an accident — and every cell on which it was previously evaluated satisfied that assumption by construction, having been balanced to a prevalence of 0.489–0.539 by the evaluation pipeline itself.

D. Cross-corpus, balanced pools

On the balanced leave-one-corpus-out protocol the prior correction is inert by construction ($\hat{\pi} \approx 0.5$ returns the symmetric budget exactly), so this arm measures the rest of the method against the train-time and test-time baselines on our own source model, ten seeds per target. Source-only EERs are 5.03 / 33.54 / 12.78 / 22.50 on ASVspoof2019 / LibriSpeech-TTS / In-the-Wild / Arabic; ours gives 3.47 / 33.74 / 11.33 / 20.99, winning on three targets (10/10 seeds on two of them, 8/10 on the third) and neutral on LibriSpeech-TTS, the target with the lowest source AUC (0.73). Per-target Wilcoxon against source-only is significant on ASVspoof2019 and Arabic ($p=0.0020$, Holm 0.0078) and nominal on In-the-Wild (0.0293). Ours beats DANN [13] and an ASDG-style baseline [5] pooled over all 40 (target, seed) pairs (30/40 each) despite DANN seeing the target’s unlabeled audio every training step. Against the modern TTA baselines on the same folds, [stage 4]. Dropping the consistency term gives 4.20 / 33.39 / 11.55 / 22.19.

Tent is unpredictably catastrophic, and sometimes silently so. On ASVspoof2019 its per-seed EER splits five seeds near source-only and four at or near chance with no warning sign (std 21.96); on Arabic it is a consistent chance-level failure (9/10 seeds). Ours shows no comparable collapse anywhere (largest std 2.01). The failure is a ranking failure first: on the strongest and weakest public checkpoints Tent collapses to chance on one (16.41 $\rightarrow$ 49.80% EER, AUC

.918 $\rightarrow$.502) but only degrades AUC on the other. On the DF pool the damage does not show in the field’s headline metric at all — Tent returns a *lower* EER than source (4.06 vs. 4.51) while saturating 98% of the pool to one tied score. A TTA study reporting EER without a score-resolution check can record a collapse as an improvement.

E. Label-free guards

An adaptation that destroys the source ranking has discarded its own precondition, and that is visible without labels: ρ , the rank agreement with the frozen source model, together with the drift of the predicted positive rate from $\hat{\pi}$ (both in Table I). Over the eight adapted DF arms they are complementary — ρ catches ranking collapse (ETA 0.31, Tent 0.44), the gap catches operating-point collapse (0.37, 0.39, 0.23) and is blind to the former — and requiring both separates the five harmful arms from the three harmless ones exactly, for any $\tau \in [0.70, 0.80]$. The plateau matters more than the exactness: an earlier ρ -only rule at $\tau=0.85$, fitted on seven arms, looked equally exact and was falsified by the eighth, rejecting the study’s best result. Two thresholds on eight points is a diagnostic, not a rule; [stage 6/7] gives the leave-one-target-out version.

F. Breadth, and the mechanism

Applied unmodified to nine released checkpoints across five architectures and ten corpora (369 cells), the consistent effect is *re-calibration*: δ falls 7.29 $\rightarrow$ 1.96 (248/369, $p=2.2 \times 10^{-17}$), and where it lands barely depends on where it started (corr=+0.17). The mechanism is *not* learned invariance: a proxy $\mathcal{A}$ -distance and a linear domain-origin probe both *rise* after adaptation (Arabic 1.545 $\rightarrow$ 1.606), the opposite of what domain-invariance predicts and consistent with DANN underperforming while optimising that very quantity. Beyond any threshold, the gradient step adds separability where the ranking is strong: on ASVspoof2019 within-attack-family AUC rises on all six families (30/30, $p=10^{-9}$), and on ITW/s42 AUC rises .917 $\rightarrow$.947; on the AUC-0.71 target it is null.

V. LIMITATIONS

The prior is not identifiable, which bounds the correction. A label-free method sees only $\Pr[\hat{y}=1] = \pi \text{TPR}_{\mathcal{T}} + (1-\pi)\text{FPR}_{\mathcal{T}}$: one equation, three unknowns. BBSE closes it by assuming source error rates carry over — what covariate shift breaks — so an extreme prior and an extreme calibration failure look identical (DF/s2 predicts 97.4% fake at a true 0.972, ITW/s42 98.1% at a true 0.372). The estimate therefore degrades exactly as the operating point does — over seven cells and two corpora, $|\hat{\pi} - \pi|$ against the source deficit gives Pearson $r=0.998$ — and a badly wrong prior is worse than none (ITW/s42 12.58 $\rightarrow$ 14.51 EER). Estimating the prior and detecting that the estimate is wrong are one problem; two label-free detectors for it are falsified in our release.

Scope. The prior fixes the *budget*, not the threshold; the gain varies (1.27 to 0.02 EER across DF checkpoints); and a low-AUC start we cannot rescue at all.

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